

HETEROSKEDASTICITY

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BE VERY WARY, VARIANCE VARIES

1.1 Nature

One of the key assumptions of a linear regression model is that the variance of the error term remains constant across all observations. When this assumption is violated, that is, the variances for all observations are not the same, we have heteroskedasticity, also known as heteroscedasticity. The random variable, y , and the random error, e , are heteroskedastic. Conversely, if all observations come from probability density functions with the same variance, homoskedasticity exists, *i.e.*, y and e are homoskedastic. When there is heteroskedasticity, the variance assumption on the least squares error term is violated. Thus, we replace the assumption of homoskedasticity with $Var(y_i) = Var(e_i) = h(x_i)$ where $h(x_i)$ is a function of x_i that increases as x_i increases.

1.2 Consequences

There are two implications of heteroskedasticity:

1. The least squares estimator is still a linear and unbiased estimator, but it is no longer best.
2. There is another estimator with a smaller variance. Thus, the standard errors computed for the least squares estimators are incorrect. Therefore, confidence intervals and hypothesis tests that use these standard errors may be misleading.

1.3 Causes

What causes heteroskedasticity?

1. **Non-constant coefficient:** Suppose that the slope coefficient varies across observations i , $y_i = \beta_1 + \beta_2 x_i + e_i$, and that it varies randomly around some fixed value α , where $\beta_2 = \alpha + u_i$. Therefore, the regression is actually $y_i = \beta_1 + \alpha x_i + u_i x_i + e_i$, where $u_i x_i + e_i$ is the error term in the regression model, *i.e.*, the error term varies with x .
2. **Omitted variables:** Suppose that the true model of y is $y_i = \beta_1 + \beta_2 x_i + \beta_3 z_i + u_i$, but the model we estimate fails to include z , then the error term in the model will capture the effect of the omitted variable, and thus, it will be correlated with z . The variance of $e_i = \beta_3 z_i + u_i$ will be non-scalar. This is known as omitted variable bias.
3. **Non-linearity:** Suppose that the true relationship is non-linear, $y_i = \beta_1 + \beta_2 x_i^2 + u_i$, but the model we estimate is linear, $y_i = \beta_1 + \beta_2 x_i + e_i$, then the residual will capture the non-linearity and its variance will be effected accordingly, $\sigma_i^2 = f(x_i^2, u_i)$.
4. **Aggregate data across groups:** When the dependent variable is constructed as an average over groups of differing sizes, then the variance of the error depends on the number of observations within each group. Larger groups produce more precise estimates of the group mean, and therefore, smaller error variances, while smaller groups produce larger error variances, resulting in heteroskedasticity.

TESTS TO DETECT HETEROSKEDASTICITY

To detect heteroskedasticity, there are numerous formal statistical tests to discuss. However, one informal way to detect heteroskedasticity is by using residual charts to plot the least squares residuals against each explanatory variable or predicted value of dependent variable.

2.1 Lagrange Multiplier Test/Breusch-Pagan Test

Consider the mean function, $\mathbb{E}[y]$, for the general multiple regression model, $\mathbb{E}[y_i] = \beta_1 + \beta_2 x_{2i} + \dots + \beta_K x_{Ki}$. A general form for the variance function is $Var(y_i) = \sigma_i^2 = \mathbb{E}[e_i^2] = h(\alpha_1 + \alpha_2 z_{2i} + \dots + \alpha_S z_{Si})$, where α 's are parameters, and z 's are explanatory variables. Two possible functions are exponential function form and linear function form. When $\alpha_2 = \alpha_3 = \dots = \alpha_S = 0$, then $h(\alpha_1 + \alpha_2 z_{2i} + \dots + \alpha_S z_{Si}) = h(\alpha_1)$, which is a constant. Thus, heteroskedasticity is not present. Essentially, we fit an auxiliary regression, *i.e.*, a linear regression of squared residuals on the explanatory variables, that is, $\hat{e}_i^2 = \alpha_1 + \alpha_2 z_{2i} + \dots + \alpha_S z_{Si} + v_i$.

The null hypothesis is $H_0 : \alpha_2 = \alpha_3 = \dots = \alpha_S = 0$ against the alternative hypothesis $H_1 : \text{Not all the } \alpha_i \text{'s in } H_0 \text{ are zero}$. Since R^2 in the auxiliary regression measures the proportion of variation in $\hat{e}_i^2$ explained by the

z 's, the test statistic is $\chi^2 = n \cdot R^2 \sim \chi_{S-1}^2$, where the sample size, n multiplied by R^2 has a χ^2 distribution with $S - 1$ d.f..

Because a large R^2 value provides evidence against the null hypothesis, the rejection region for the statistic is in the right tail of the distribution. For a 5% significance level, we reject H_0 and conclude that heteroskedasticity exists when $\chi^2 \geq \chi_{(0.95; S-1)}^2$

The advantages of a Lagrange multiplier test/Breusch-Pagan test is that the value of the statistic computed from the linear function is valid for testing an alternative hypothesis of heteroskedasticity where the variance function can be of any form. However, the disadvantages are that it is a large sample test, and the results hold approximately in large samples. You also need to know all the variables that cause the heteroskedasticity.

2.2 The White Test

As mentioned in the disadvantages of the Breusch-Pagan test, we assume that we know all the relevant variables, z 's. In practice, we may wish to test for heteroskedasticity without precise knowledge of the relevant variables. Hal White, in 1980, suggested defining the z 's as equal to the x 's, the squares of the x 's, and possibly their cross-products. Suppose the mean function has two explanatory variables, $\mathbb{E}[y] = \beta_1 + \beta_2 x_2 + \beta_3 x_3$. The White test without cross-product interactions specifies that $z_2 = x_2, z_3 = x_3, z_4 = x_2^2, z_5 = x_3^2$, and including interactions adds another variable, $z_6 = x_2 x_3$. The White test is performed as an F -test or using $\chi^2 = n \cdot R^2$.

The advantages of a White test are that the test does not make any assumptions about the particular form of heteroskedasticity, and so, is quite general in application. It also does not require that the error terms be normally distributed. However, some disadvantages are, rejecting the null may be an indication of model specification error, as well as or instead of heteroskedasticity. It might reveal heteroskedasticity, but it might also simply be rejected as a result of missing variables. White's standard errors have been found to be unreliable in small samples but revised methods for small samples have been developed to allow robust standard errors to be calculated for small n .

2.3 The Goldfeld-Quandt Test

The Goldfeld-Quandt test is designed to test for two groups of data with possibly different variances. Goldfeld and Quandt, in 1965, suggested the following test procedure for null and alternative hypotheses of the form, $H_0 : \sigma_1^2 = \sigma_2^2$ against $H_0 : \sigma_1^2 \neq \sigma_2^2$. We order the observations in ascending order of x , and omit ρ central observations ($\rho \approx \frac{n}{3}$). Fit the separate regression to both sets of observations, $F = \frac{\hat{\sigma}_2^2 / \sigma_2^2}{\hat{\sigma}_1^2 / \sigma_1^2} \sim F_{(n_2 - K_2, n_1 - K_1)}$. Note that it is convenient if you always put the larger estimate in the numerator, then you can reject H_0 at a 5% level of significance if $F > F_c = F_{(0.975; n_2 - K_2, n_1 - K_1)}$, where F_c is the critical value in the upper tail.

Although the Goldfeld-Quandt test is specifically for the cases where the sample divides naturally into two groups, it can also be used where, under H_1 , the variance is a function of a single explanatory variable. To do that, we order the observations according to that single explanatory variable, then split sample into approximately two halves, carry out two separate least squares regression that yield variance estimates, and proceed with the Goldfeld-Quandt test procedures as described above.

METHODS TO CORRECT HETEROSKEDASTICITY

3.1 Known Form of Variance

One method to correct heteroskedasticity, with known form of variance, is the **White's heteroskedasticity-consistent standard errors**, also known as robust standard errors. In the variance for a simple linear regression model, we replace σ_i^2 with the least squares residuals $\hat{e}_i^2$ and introduce a degrees of freedom adjustment $\frac{n}{n-K}$, thus, the White variance estimator for simple regression is $\widehat{Var}(b_2) = \frac{n}{n-2} \frac{\sum [(x_i - \bar{x})^2 \hat{e}_i^2]}{[\sum (x_i - \bar{x})^2]^2}$, and the White standard error is given by the square root of this quantity.

The White's estimator for the standard errors helps avoid computing incorrect interval estimates or incorrect values for test statistics in the presence of heteroskedasticity. However, the White's correction does not address the other implication of heteroskedasticity, that is, the least squares estimator is no longer best. Failing to address this issue may not be serious, with a large sample size, the variance of the least squares estimator may still be sufficiently small to get precise estimates. To find an alternative estimator with a lower variance is necessary to specify a suitable variance function, while using least squares with robust standard errors avoids the need to specify a suitable variance function.

To develop an estimator that is better than the least squares estimator we need to make a further assumption about how the variance, σ_i^2 , changes with each observation, although it is possible to obtain the White corrected variance estimates. An estimator known as the **generalised least squares estimator** depends on σ_i^2 , which is

unknown. To make the generalised least squares estimator operational, some function is imposed on σ_i^2 . For example, one possibility is that $Var(e_i) = \sigma_i^2 = \sigma^2 x_i$. We now can transform the model into one with homoskedastic errors, $\frac{y_i}{\sqrt{x_i}} = \beta_1 \left(\frac{1}{\sqrt{x_i}} \right) + \beta_2 \left(\frac{x_i}{\sqrt{x_i}} \right) + \frac{e_i}{\sqrt{x_i}}$, defining the transformed variables as $y_i^* = \frac{y_i}{\sqrt{x_i}}$, $x_{1i}^* = \frac{1}{\sqrt{x_i}}$, $x_{2i}^* = \frac{x_i}{\sqrt{x_i}} = \sqrt{x_i}$, $e_i^* = \frac{e_i}{\sqrt{x_i}}$. Thus, the transformed model is $y_i^* = \beta_1 x_{1i}^* + \beta_2 x_{2i}^* + e_i^*$, where $Var(e_i^*) = \sigma^2$. The estimators are the BLUE. Do note that the wrong choice of weights can produce biased estimates of the standard errors, and we can never know for sure whether we have chosen the correct weights.

3.2 Unknown Form of Variance

With unknown form of variance, we can still use the **generalised least squares estimator** method. We can transform the variables by $\hat{\sigma}_i$ and continue as described above. Robust standard errors can be used not only to guard against the possible presence of heteroskedasticity when using least squares, but also used to guard against the possible misspecification of a variance function when using generalised least squares.

CONCLUSION

Heteroskedasticity is a common problem in applied regression analysis, particularly when working with aggregated data. Whilst OLS estimators remain unbiased and consistent in its presence, it is no longer best, the distortion of standard errors leads to unreliable inference. Consequently, it is important to understand the sources of heteroskedasticity, recognise its effects on regression results, and apply appropriate tests. When heteroskedasticity is detected, corrective measures such as robust standard errors, variable transformations, weighted least squares, *etc.* can help restore the reliability of statistical inference and improve the usefulness of the regression model.

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